

Q30

$$A = \lambda N = \left(\frac{\ln 2}{t_{\frac{1}{2}}} \right) \left(\frac{M}{M_m} N_A \right) \text{ or}$$

$$M = \frac{A t_{\frac{1}{2}} M_m}{(\ln 2) N_A} = \frac{(3.7 \times 10^{10})(138 \times 24 \times 60 \times 60)(210)}{(\ln 2)(6.02 \times 10^{23})} = 2.2 \times 10^{-4} \text{ g}$$

= 0.22 mg

Ans: A

***** END *****